

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (CL) Examination****May 2018****CL308 Geotechnical Engineering - II****Date: 05.05.2018, Saturday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use of IS: 6403 -1981 is allowed.

SECTION – I

- Q - 1 (a)** Enlist different kind of soil boring methods and describe any two methods briefly. **[04]**
- (b)** Explain different type of corrections applied to field N values. **[05]**
- Q - 2 (a)** Explain earth pressure at rest condition. Derive equation for co-efficient of earth pressure at rest condition (K_0). **[06]**

OR

- (a)** Enumerate steps to calculate active earth pressure graphically with Culmann's method. **[06]**
- (b)** A vertical excavation was made in a clay having unit weight of 20 kN/m^3 . It caved in after the depth of digging reaches 4 m. Taking the angle of internal friction to be zero, calculate the value of cohesion. If the same clay is used as a backfill against a retaining wall, up to a height of 8 m, calculate active earth pressure force before development of crack on surface. **[07]**
- Q - 3 (a)** In order to find factor of safety of downstream slope of an earthen dam during steady seepage condition, section of the dam was drawn to a scale of 1 cm = 3 m. Following results were obtained from the analysis: **[03]**
- $\sum N = 620 \text{ kN}$
 $\sum T = 440 \text{ kN}$
 $\sum U = 150 \text{ kN}$
 Length of arc is 16 cm.
 Laboratory tests have furnished values of $\phi = 30^\circ$ and $c = 25 \text{ kN/m}^2$. Unit weight of soil is 18 kN/m^3 . Determine factor of safety of the slope.
- (b)** Determine factor of safety against sliding along a slip surface for an earthen slope of soil having $c = 55 \text{ kN/m}^2$ and $\phi = 0^\circ$. Radius of slip circle is 19 m. Weight of failure wedge is 2050 kN. Perpendicular distance between C.G. of failure wedge and center of slip circle is 9 m. Angle subtended by arc at center is 64.5° . **[04]**
- (c)** How stability of finite slope is determined with Taylor's stability number? **[06]**

OR

- (c)** Derive equation of factor of safety for infinite slope of cohesionless soil. **[06]**

SECTION – II

- Q - 4 (a)** A point load of 400 kN is acting on the surface of a soil mass. Find vertical stress at a depth 2 m for different horizontal distance 0 m, 1 m, 2 m and 3 m from the axis of the load by Boussinesq's solution. [04]
- (b)** A column of a building transfers a concentrated load of 1200 kN to the soil in contact with the footing. Estimate vertical stress at 1 m & 3 m vertically below the column load by making use of the Westergaard's equation. [04]

OR

- (b)** What is Newmark's influence chart? For determination of vertical stress of $0.1q$, $0.2q$ and $0.3q$ at 5 m depth, find out radius of first three circles of Newmark's chart. 'q' is intensity of loading for circular area. Assume suitable scale and number of sectors for each circle. [04]
- Q - 5 (a)** Explain different criteria for deciding depth of foundation. [06]
- (b)** Calculate net safe load for square footing of 3 m width founded at depth of 2.5 m in $c - \phi$ soil having $c = 37 \text{ kN/m}^2$ and $\phi = 27^\circ$. Water table is at a depth of 0.75 m from the ground surface. Saturated and dry unit weight of soil is 20 kN/m^3 and 16 kN/m^3 respectively. Assume local shear failure and use IS code method. Take FOS = 3. [07]

OR

- (b)** Calculate gross safe bearing capacity of strip footing 2 m wide founded at a depth of 1.2 m in sand assuming general shear failure. Water table is at a depth of 0.5 m from the ground surface. The saturated unit weight of sand is 19.5 kN/m^3 and unit weight above water table is 16.8 kN/m^3 . The shear parameters are $c = 0$ and $\phi = 35^\circ$. Take FOS = 2.5. Take $N_c = 57.8$, $N_q = 41.4$ and $N_\gamma = 42.4$. Use Terzaghi's bearing capacity theory. [07]
- Q - 6 (a)** State & Describe various dynamic formulas used for estimating pile bearing capacity. Enumerate their limitations. [03]
- (b)** Discuss in detail the "Negative skin friction of pile". [04]
- (c)** A square pile is founded in medium dense sand having angle of internal friction 32° . Pile is having side 0.3 m X 0.3 m and 10 m in length. Calculate allowable load capacity of pile. Take unit weight of sand 18 kN/m^3 , consider critical depth $L_{cr} = 15B$. For $\phi = 32^\circ$, $N_q = 27$ and $N_\gamma = 4.1$ (Use Terzaghi's approach). Pile friction angle $\delta = 0.75\phi$, $K_o = 1$ and factor of safety as 2.5. [07]

OR

- (c)** Square pile group of 16 pile is embedded in clay soil. The diameter of each pile is 0.25 m and center to center spacing between pile is 0.6 m. If $c = 65 \text{ kN/m}^2$, determine whether the failure would occur with pile acting individual or as a group. All piles are 12 m long and take $\alpha = 0.75$. Also determine the efficiency of group pile. [07]
